`repair` Package Outline

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Package Plan

The `repair` package will fit regression models among random effect predictions using the REPAIR method (Vig & Hallquist, in prep). It is intended to be "one size fits all" in that both level 1 (MLM) and level 2 (regression) models can be specified and fit in a single function call. Summary and print methods will provide appropriate information formatted similarly to `lm()`, `lmer()`, and `lavaan()` outputs. Internally, the package will use an OOP approach via S7 classes and methods (superseding the S4 implementation in `lme4` and `lavaan`).

Formula Approach

Both level 1 and level 2 models will be fit using native R formulas. Level 1 models will follow the regular `lme4` syntax:

```
f1 <- y ~ 1 + x + (1 + x|id)
```

Level 2 models will introduce a new, but simple, syntax via the `re()` helper (for random effect). This small wrapper will specify the random effect from the first model the user wishes to use as a variable, e.g.:

```
f2 <- w ~ 1 + re("x")
```

where `re("x")` is the random slope from `f1`, grouped by `id`. Random intercepts will be specified using `re("1")`, or `re("Intercept")`.

Main `repair` Function

The workhorse function of the package will simply be `repair()`. It will take arguments specifying one (or more) level 1 formulas, a level 2 formula (possibly more), and data for each level. Multiple formulas and datasets will be provided as lists and can be named. The output of the function will be of class `repairMod`

Steps

The rough internal steps of the function will be:

1. Validate input types
2. Parse formulas
3. Compare formulas to data and ensure all variables are present/not duplicated
4. Run level 1 models via `lmer` or `lme`
5. Run Fuller's M-MOM for level 2 models
6. Gather estimates and diagnostics
7. Create `repairMod` class object and return